

Data Mining Lecture 7 – Classification Part 1

1-Classification is a data analysis task where a model is constructed to predict class labels .

- a- True
- b- False

Answer : a.

2-What is the first step in the two-step process of constructing a classification model ?

- a- Testing
- b- Validation
- c- Optimization
- d- Learning

Answer : d.

3- What is done during the learning (training) step of constructing a classification model?

- a- Predict class labels for given data
- b- Validate the model's performance
- c- Build a classifier for a predetermined set of classes
- d- Optimize the model parameters

Answer : c.

4- What is the purpose of the classification step in constructing a classification model?

- a- Building the classifier
- b- Optimizing the model parameters
- c- Predicting class labels for given data
- d- Evaluating the model's performance

Answer : c. Note “ classification is associated with supervised learning “

5- What is decision tree induction?

- a- Learning from a training dataset using neural networks
- b- Learning from a training dataset using decision trees
- c- Learning from a training dataset using support vector machines
- d- Learning from a training dataset using k-nearest neighbors

Answer : b.

6-What does an internal node represent in a decision tree?

- a- A class label
- b- A test on an attribute
- c- A test outcome
- d- A branch

Answer : b.

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7-In a decision tree, what does a branch represent?

- a- A class label
- b- A test on an attribute
- c- A test outcome
- d- An internal node

Answer : c.

8- What does a leaf node represent in a decision tree?

- a- A class label
- b- A test on an attribute
- c- A test outcome
- d- An internal node

Answer : a.

9- Decision tree induction is particularly advantageous because:

- a- It requires extensive domain knowledge
- b- It needs parameter setting for optimal performance
- c- It can handle only one-dimensional data
- d- It can handle multidimensional data without difficulty

Answer : d.

10- Decision tree could be used with _____ data.

- a- Discrete
- b- Continuous
- c- Both
- d- None

Answer : c.

11- What is the purpose of the Splitting Criterion in decision tree induction?

- a- To determine the size of the decision tree
- b- To decide which attributes to test at each node
- c- To assign class labels to leaf nodes
- d- To calculate the impurity of the dataset

Answer : b.

12- In decision tree induction, what does it mean for a partition to be "pure"?

- a- All tuples in the partition belong to different classes
- b- All tuples in the partition belong to the same class
- c- The partition contains an equal number of tuples from each class
- d- The partition contains tuples from all possible classes

Answer : b.

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13- What happens if there are no tuples for a given branch in decision tree induction?

- a- The branch is pruned from the tree
- b- A new attribute is chosen to split the data
- c- A leaf node is created with the majority class in the data partition
- d- The tree is terminated

Answer : c.

14- Which of the following is NOT an attribute selection measure used in decision tree induction?

- a- Information Gain
- b- Gain Ratio
- c- Alaadin Gini
- d- Gini Index

Answer : c.

15- Attribute selection measures rank attributes according to a measure, and the attribute with the best score is chosen as the splitting attribute.

- a- True
- b- False

Answer : a.

16- What is the goal of Information Gain in decision tree induction?

- a- To maximize the expected number of tests needed to classify a tuple
- b- To minimize the information needed to classify tuples in resulting partitions
- c- To maximize the impurity in resulting partitions
- d- To maximize the number of attributes used for splitting

Answer : b.

17- Gain Ratio is a measure used in decision tree induction that guarantees a simple tree is found.

- a- True
- b- False

Answer : b, explanation “ Information Gain instead of gain ratio “

18- Which measure used in decision tree induction is based on Shannon's information theory?

- a- Information Gain
- b- Gain Ratio
- c- Gini Index
- d- None of the above

answer : a.

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19-The smaller the expected information still required, the lower the purity of the partitions.

- a- True
- b- False

Answer : b, explanation “ The smaller the expected information still required, the greater the purity of the partitions “

20- What does Gain(A) represent in decision tree induction?

- a- The difference between the original information required and the new requirement after partitioning on A
- b- The total information gain of attribute A
- c- The entropy of the dataset D
- d- The proportion of classes in the dataset D

Answer : a, note “ that’s the Information gain “

21- How is the splitting attribute chosen in decision tree induction?

- a- The attribute with the lowest Gain(A)
- b- The attribute with the highest Gain(A)
- c- The attribute with the highest entropy
- d- The attribute with the lowest entropy

Answer : b.

department	age	salary	status	count
sales	Middle aged	medium	senior	30
sales	youth	low	junior	30
sales	Middle aged	low	junior	40
systems	youth	medium	junior	20
systems	Middle aged	high	senior	20
systems	senior	high	senior	10
marketing	senior	medium	senior	10
marketing	Middle aged	medium	junior	20
secretary	senior	medium	senior	10
secretary	youth	low	junior	10

$$info_A(D) = \sum_{j=1}^v \frac{|D_j|}{D} \times info(D_j)$$

$$Info(D) = \text{entropy} = - \sum_{i=1}^m p_i \log_2(p_i)$$

$$Gain(A) = info(D) - info_A(D)$$

Based on the following data .

22- In decision tree induction, if a dataset contains 100 tuples of Class 1 (Senior) and 150 tuples of Class 2 (Junior), what is the entropy (Info(D)) of the dataset?

- a- 0.67
- b- 0.92
- c- 0.97
- d- 1.09

$$\begin{aligned} info(D) = \text{entropy} &= - \sum_{i=1}^m p_i \log_2 p_i \\ &= - \frac{80}{200} \log_2 \frac{80}{200} - \frac{120}{200} \log_2 \frac{120}{200} = 0.97 \end{aligned}$$

Answer : c.

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23- What is the purpose of tree pruning in decision tree induction?

- a- To increase the complexity of the decision tree
- b- To remove the most reliable branches
- c- To remove the least reliable branches and simplify the tree
- d- To introduce anomalies and outliers into the dataset

Answer : c.

24- Which of the following statements is true about prepruning in decision tree induction?

- a- It removes sub-trees from already constructed trees.
- b- It statistically assesses the goodness of a split after it takes place.
- c- It is easy to choose thresholds for statistical significance.
- d- It assesses the goodness of a split before it takes place.

Answer : d.

25- What is the main drawback of prepruning in decision tree induction?

- a- It leads to underfitting of the data.
- b- It requires hard thresholds for statistical significance.
- c- It increases the complexity of the decision tree.
- d- It removes the most reliable branches.

Answer : b.

26- In postpruning, how are sub-trees removed from already constructed trees?

- a- By adding leaf nodes
- b- By merging with other sub-trees
- c- By splitting into smaller sub-trees
- d- By removing sub-tree branches and replacing with a leaf node

Answer : d.

27- What determines the label of the leaf node in postpruning?

- a- The least frequent class in the sub-tree
- b- The class with the highest entropy in the sub-tree
- c- The class with the least reliability in the sub-tree
- d- The most frequent class in the sub-tree

Answer : d.

28- When does rules conflict occur in decision tree induction?

- a- When a tuple fires multiple rules with the same class prediction.
- b- When a tuple fires a rule with different class predictions.
- c- When there are no rules available for a tuple.
- d- When a tuple is incorrectly classified by the decision tree.

Answer : a.

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29- Size ordering strategy in resolving rule conflicts prioritizes rules based on the size of their antecedents.

a- True

b- False

Answer : b. explanation “ rule ordering”

30- In rule-based ordering for resolving rule conflicts, what measures can be used to prioritize rules?

a- Accuracy

b- Size

c- Domain expertise

d- All of the above

Answer : d.

31- Fallback (default) rule is applied when no rules are triggered for a tuple in decision tree induction.

a- True

b- False

Answer : a.

" شكرا لكم ولا تنسونا بالدعاء "